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Title: The effect of seeing scientists as intellectually humble on trust in scientists and their research

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Abstract

Public trust in scientists is critical to our ability to face societal threats. Across five preregistered studies (N=2,034), we assessed whether perceptions of scientists' intellectual humility affect perceived trustworthiness of scientists and their research. In Study 1, we found that seeing scientists as higher in intellectual humility was associated with greater perceived trustworthiness of scientists and support for science-based beliefs. We then demonstrated that describing a scientist as high (vs. low) in intellectual humility increased perceived trustworthiness of the scientist (Studies 2-4), belief in their research (Studies 2-4), intentions to follow their research-based recommendations (Study 3), and information-seeking behavior (Study 4). We further demonstrated that these effects were not moderated by the scientist's gender (Study 3) or race/ethnicity (Study 4). In Study 5, we experimentally tested communication approaches scientists can use to convey intellectual humility. These studies reveal the benefits of seeing scientists as intellectually humble across medical, psychological, and climate science topics.

Main Text

Public trust in science and scientists is critically important to scientific endeavors and evidence-based social policy. While trust in science and scientists is generally high¹, in 2021 the percentage of Americans who reported a great deal of confidence in scientists dropped to 29% (a 10% decrease from 2020)². Lower levels of trust are more prevalent for polarizing topics and among certain groups of the population (e.g., American conservatives),^{3,4} indicating heterogeneity in how people perceive scientists and their recommendations. During the COVID-19 pandemic, for instance, we witnessed the deleterious effects that lower trust can have on people's tendency to follow public health guidelines. People with lower levels of trust in science reported weaker intentions to comply with recommendations to socially distance⁵ despite social distancing being identified as one of the most effective practices for stopping the spread of COVID-19⁶. Moreover, evidence from 12 countries demonstrated that people with lower trust in science are less likely to vaccinate against COVID-19⁷. Looking beyond the COVID-19 pandemic, lower trust in science decreases belief in threats such as climate change⁸ and reduces people's willingness to engage in pro-environmental behaviors⁹. As such, lower trust in science often has personal and societal costs and can even result in the loss of human lives.

Given the costs of lower trust in science and scientists, it is important that we understand its origins. Researchers have identified many separable but overlapping paths that can lead to lower trust in science^{10,11}. Certain political and religious beliefs can undermine trust in science¹⁰⁻¹⁵, and people sometimes have lower trust in scientists because they believe scientific findings or values conflict with their own worldviews¹⁶⁻¹⁸. People may even assume that scientists are not primarily focused on finding objective truths but instead are biased and politically motivated¹⁹. Other belief systems and worldviews, including conspiracy beliefs, can also give rise to lower trust in science and advance suspicions that scientists do not have the public's best interests at heart or are lying to the public²⁰⁻²².

People may also have lower trust in science because of the shortcomings and uncertainties inherent in the scientific process. Namely, new and socially relevant scientific evidence is often tentative, uncertain, or conflicting²³. Scientific research understandably cannot reach complete or unequivocal conclusions because of the complexity of what is being studied, contextual influences, or measurement error (among other reasons)^{24,25}. Scientists themselves are also limited. Scientists can make mistakes or incorrect predictions^{26,27} and can develop methods or theories that perpetuate prejudice and bias²⁸. Further, although the scientific process provides a safeguard against confirmation bias, "researcher degrees of freedom" can lead even the most well-intentioned scientists to conduct different statistical analyses, leading to different results and scientific conclusions²⁹, ultimately resulting in broader problems such as the replication crisis. Therefore, blind trust in all scientific findings is not ideal and remaining skeptical is an important feature of scientific thinking. Nevertheless, considering science as a trustworthy source of knowledge is fundamental for evidence-based thinking, learning, and information evaluation for both scientists and non-scientists³⁰.

1 Notwithstanding valid reasons for qualified trust in science and scientists, the aim of the
 2 present work is to understand factors that legitimately promote or hinder trust. Prior research
 3 makes the distinction between trust as a perception (e.g., seeing a scientist as worthy of being
 4 trusted) and as a behavior (e.g., following a scientist's recommendations)³¹. In the current
 5 research we were interested in perceived trustworthiness—conceptualized as having the qualities
 6 of expertise (i.e., seeing the scientist as epistemically competent), benevolence (i.e., seeing the
 7 scientist as concerned for the wellbeing of others), and integrity (i.e., seeing the scientist as
 8 honest)³²—as well as whether this perceived trust spills over into actual behavioral willingness to
 9 follow recommendations.

10 We argue that certain features of scientists might exert powerful influences on how
 11 trustworthy they appear to be. Recently, scholars^{33,34} have called for scientists to act and
 12 communicate with greater intellectual humility (IH)—the awareness that one's knowledge and
 13 beliefs might be limited or wrong³⁵. While IH is most often defined as an intrapersonal or meta-
 14 cognitive trait, other definitions include interpersonal and context-specific manifestations^{35,36}. In
 15 the current research, we similarly define IH as the intrapersonal awareness of the limitations of
 16 one's knowledge, which can in turn have interpersonal and perceivable aspects (e.g.,
 17 respectfulness). IH is distinct from other similar traits, such as general humility, in that it
 18 specifically focuses on the limitations of one's knowledge, as opposed to one's general
 19 limitations^{35,37}. In this way, IH is uniquely relevant to the domain of science relative to general
 20 humility, as being aware of the limitations inherent to the scientific enterprise and
 21 communicating transparently about them is critical for improving the scientific process³³. As
 22 such, IH may be perceived as the expression of following the norms of science³³. Because
 23 scientific norms so closely resemble what it means to be intellectually humble, when scientists
 24 fail to behave in ways that reflect intellectual humility, it might be especially detrimental and
 25 jarring, as it goes against both the fundamental norms of science and people's expectations for
 26 how a responsible scientist should act. Just because scientists understand that the scientific
 27 process entails these features, they may not always behave in ways that reflect IH. We therefore
 28 ask: how do people's perceptions of scientists' IH influence their perceived trustworthiness of
 29 scientists and willingness to follow research-based recommendations?

30 To our knowledge, no one has yet examined the impact of perceptions of scientists' IH on
 31 perceived trustworthiness of them and their research. Prior work has examined how someone's
 32 trait levels of IH predict their trust in science^{38,39}, and how a scientist's communication of topic
 33 complexity can impact someone's trust and levels of IH about a scientific topic⁴⁰. Other work has
 34 demonstrated that certain behavioral qualities of scientists that might map onto high IH are
 35 associated with positive outcomes. For example, scientists who admit to mistakes and support
 36 open science practices are perceived as more trustworthy^{41,42}, and expressing uncertainty and
 37 complexity as part of scientific communication can sometimes foster trust^{40,43,44}. While these
 38 studies focus on specific methods of conducting or communicating about science, they do not
 39 directly assess more global perceptions of scientists' IH. One study on physician humility
 40 demonstrated that participants reported greater trust and willingness to adhere to the

recommendations of fictitious physicians who were ostensibly described by a prior patient as “very humble (vs. “not humble”) in [their] approach to patient care”⁴⁵. These studies, however, focused on patients’ perceptions of physicians’ humility toward their patients and did not assess characteristics of IH or perceived trustworthiness of scientists and their research.

We argue that perceptions of scientists’ IH may be an especially important predictor of people’s perceived trustworthiness of scientists because IH uniquely focuses on admitting the limitations of one’s knowledge and beliefs. Because science is inherently uncertain and limited, being willing to recognize when one is wrong, rely on others’ knowledge, or update one’s beliefs, is critical for the scientific process³³. We therefore hypothesized that these characteristics of IH render it particularly relevant to the perceived trustworthiness of scientists.

More specifically, we reasoned that perceiving a scientist as having low IH should feed into lower perceived trustworthiness because being unwilling to recognize one’s fallibility can decrease the quality of one’s research. That is, low IH may impede aspects of the scientific process that make it valuable and rigorous (e.g., avoiding confirmation bias; separating one’s ego from one’s research program; being willing to update beliefs based on data; recognizing when one has made an error; recognizing limitations of the science; continuously learning; leaning on the knowledge of others). Indeed, people who are lower in IH are less scrutinizing of information that confirms their worldviews⁴⁶, are less effective at recognizing gaps in their knowledge⁴⁷, are more overconfident in their knowledge⁴⁸, are less motivated by learning⁴⁹, use less effective learning strategies⁵⁰, and are less open to exposing themselves to different perspectives⁵¹. Although it may be difficult for participants to understand how low IH is related to specific questionable scientific practices, scientists who express an insight into the limits of their own knowledge and their continuous learning and updating of beliefs would be perceived as trustworthy in their general approach to research and science⁴¹.

By contrast, we reasoned that perceiving a scientist as having higher (vs. lower) IH should foster perceived trustworthiness because being willing to recognize one’s fallibility supports learning-oriented approaches and rigorous scientific practices like those listed above, leading to perceptions of expertise. Moreover, because people higher (vs. lower) in IH are more likely to endorse prosocial values⁵² and are perceived as having more prosocial characteristics^{53,54}, people might believe that scientists with higher IH have the public’s best interests at heart, leading to perceptions of benevolence. Finally, IH may also imply that the scientist is less motivated by personal or political agendas because they are more focused on pursuing knowledge, leading to perceptions of integrity. We therefore predicted that people would report greater perceived trustworthiness of scientists—in the form of greater perceived expertise, benevolence, and integrity—who they perceive as having high (vs. low) IH. Further, we aimed to examine whether, in addition to perceiving scientists as trustworthy, participants would exhibit more behavioral trust—operationalized as seeing the scientist’s research findings as trustworthy and being interested in receiving more information about how to follow their research-based recommendations.

Overview

In the current work, we assessed perceptions and outcomes of seeing scientists as intellectually humble in five preregistered studies (N=2,034). In Study 1, we examined the correlates of seeing scientists as intellectually humble and found that those who saw scientists as higher in IH indicated higher perceived trustworthiness of scientists and support for science-based beliefs. In Study 2, we experimentally manipulated the IH of an ostensible scientist, demonstrating that describing a scientist as high (vs. low) in intellectual humility increases their perceived trustworthiness and trust in their research findings. In Studies 3 and 4, we replicated this finding while also testing for moderation by the gender (Study 3) and race/ethnicity (Study 4) of the scientist, and further demonstrated effects of a scientist's IH on intentions to follow their research-based recommendations (Study 3) and information-seeking behavior aimed at learning more about the scientist's recommendations (Study 4). In a final study, we tested tangible communication approaches scientists might use to convey their IH to the public in a census-matched representative experiment. In all studies, we show the benefits of seeing scientists as intellectually humble across a variety of polarized scientific domains including medical science (Studies 1 and 2), psychological science (Studies 3 and 5), and climate science (Studies 1 and 4). For all studies, we also present exploratory Bayes Factors for null effects using JASP version 0.18.3 and JASP's default priors^{58,59}. Bayesian results did not substantially differ when testing robustness with different priors (see OSF). All descriptive information and plots of posterior distributions are available on OSF.

While not preregistered, we also explored the influence of perceived IH on each dimension of perceived trustworthiness (i.e., expertise, benevolence, integrity), as measured by the Muenster Epistemic Trustworthiness Inventory³². Further, while there are many IH scales that conceptualize IH in slightly different ways, in the current work we focus on the conceptualization of IH captured by the Comprehensive Intellectual Humility Scale^{36,55}. We chose this scale because it contains subscales that assess both the more intrapersonal dimensions of IH (lack of intellectual overconfidence, independence of intellect and ego) as well as subscales that have some interpersonal components and are therefore more observable (openness to revising one's views, respect for others' views). Prior work shows that this scale shows similar associations with various outcomes as other IH scales that focus on purely meta-cognitive aspects^{56,57}. While this scale is not normally used to measure perceptions of someone else's IH, throughout all studies we found that the adapted version of the scale was affected by our manipulations of IH, suggesting that it was picking up on signals of scientists' IH as we had intended it to. We present exploratory analyses using the separate IH subscales here and in the supplementary materials (SM).

Results

Study 1

Study 1 was a correlational test of the association between seeing scientists as intellectually humble and trustworthy. We asked participants (N = 298) to think of a broad range of scientists and rate them on perceived IH⁵⁵. Participants also provided ratings of their perceived

trustworthiness of scientists³² and indicated their belief in polarizing science topics (climate change, vaccinations, genetically modified foods)²¹.

Perceived IH of scientists was strongly associated with perceived trustworthiness of scientists ($r(296) = .62, p < .001$) and belief in anthropogenic climate change ($r(296) = .43, p < .001$), support for vaccinations ($r(296) = .49, p < .001$), and support for genetically modified food ($r(296) = .43, p < .001$; see Figure 1), with effect sizes ranging from medium-large to large⁶⁰. While all subscales of IH of scientists were associated with perceived trustworthiness, they varied in magnitude (openness to revising one's views $r(296) = .54, p < .001$; respect for other's views $r(296) = .62, p < .001$; lack of intellectual overconfidence $r(296) = .46, p < .001$; independence of intellect and ego $r(296) = .43, p < .001$). Similarly, while IH of scientists was associated with all three dimensions of perceived trustworthiness, they varied in magnitude (integrity $r(296) = .62, p < .001$; benevolence $r(296) = .57, p < .001$; expertise $r(296) = .47, p < .001$). Importantly, perceived IH of scientists did differ by political orientation with a large effect size, with liberal participants perceiving more IH in scientists ($M = 3.74, SD = 0.51$) than conservative participants ($M = 3.15, SD = 0.70; F(1, 296) = 70.24, p < .001, \eta^2_p = .19, 95\% CI$ for $\eta^2_p = (.12, .27)$; see Figure 2). However, associations between perceived IH of scientists and perceived trustworthiness, and between perceived IH of scientists and belief in polarizing science topics, were robust to including simultaneous covariates of political orientation, political conviction, interest in science, science knowledge, and conspiracy mentality ($ps < .021$; see Table S3). Finally, perceived IH of scientists was still significant when also including cognitive reflection and participants' own level of IH to these models ($ps < .001$), with the exception of the belief in climate change model ($p = .083$).

Overall, Study 1 provided correlational evidence that people who see scientists as intellectually humble also tend to see those scientists as more trustworthy and hold evidence-based scientific beliefs.

Study 2

In Study 2, we tested our hypothesis using an experimental design. We randomly assigned participants ($N = 317$) to read one of three articles about an ostensible scientist, Dr. Susan Moore, researching new treatments for long COVID-19 symptoms. Participants in the control condition read about Dr. Susan Moore's research and findings. Participants in the low IH condition read this same article but with cues of low IH (e.g., "...according to one colleague, 'Dr. Moore is not afraid to assert what she knows'"). Finally, participants in the high IH condition read the same control article but with cues of high IH (e.g., "...according to one colleague, 'Dr. Moore is not afraid to admit when she doesn't yet know something'"; see OSF for manipulations).

We found a large effect of our manipulation on perceived IH of the scientist, $F(2, 314) = 172.84, p < .001, \eta^2_p = .52, 95\% CI$ for $\eta^2_p = (.45, .58)$. Using a Tukey test to adjust for multiple comparisons, participants saw the scientist as significantly more intellectually humble in the high IH condition ($M = 4.19, SD = 0.57$), than in either of the other conditions ($M_{\text{Control condition}} = 3.61, SD_{\text{Control condition}} = 0.51; M_{\text{Low IH condition}} = 2.70, SD_{\text{Low IH condition}} = 0.69; p's < .001$). Participants saw

the scientist as significantly less intellectually humble in the low IH condition than in the control condition ($p < .001$). The strength of the effect of condition varied by subscale of IH (lack of intellectual overconfidence $\eta^2_p = .57$, 95% CI for $\eta^2_p = (.51, .63)$; openness to revising one's views $\eta^2_p = .51$, 95% CI for $\eta^2_p = (.44, .57)$; respect for other's views $\eta^2_p = .31$, 95% CI for $\eta^2_p = (.23, .38)$; independence of intellect and ego $\eta^2_p = .25$, 95% CI for $\eta^2_p = (.17, .33)$).

We also found a large effect on perceived trustworthiness, $F(2, 314) = 25.50$, $p < .001$, $\eta^2_p = .14$, 95% CI for $\eta^2_p = (.07, .21)$ (see Figure 3). Participants saw the scientist as significantly more trustworthy in the high IH condition ($M = 6.43$, $SD = 0.64$), than in the low IH condition ($M = 5.62$, $SD = 1.10$; $p < .001$). Participants saw the scientists as significantly less trustworthy in the low IH condition than in the control condition ($M = 6.17$, $SD = 0.76$; $p < .001$). The difference between the high IH and control conditions did not reach significance ($p = .078$), though a Bayes factor analysis indicated some evidence against the null (that group means do not differ; $BF_{01} = 0.27$). The strength of the effect of condition differed by perceived trustworthiness subscale (integrity $\eta^2_p = .13$, 95% CI for $\eta^2_p = (.07, .20)$; benevolence $\eta^2_p = .19$, 95% CI for $\eta^2_p = (.11, .26)$; expertise $\eta^2_p = .06$, 95% CI for $\eta^2_p = (.02, .12)$).

Finally, we found a small-to-medium effect on belief in the scientist's research on the new long COVID-19 treatment, $F(2, 314) = 6.56$, $p = .002$, $\eta^2_p = .04$, 95% CI for $\eta^2_p = (.01, .09)$. Participants believed the scientist's research more in the high IH condition ($M = 4.85$, $SD = 1.07$), than in the low IH condition ($M = 4.45$, $SD = 1.14$; $p = .025$). Participants believed the scientist's research less in the low IH condition than in the control condition ($M = 5.00$, $SD = 1.16$; $p = .002$). The difference between the high IH and control condition did not reach significance ($p = .589$), and the Bayes factor provided evidence in favor of the null hypothesis ($BF_{01} = 4.23$).

All results remained significant after controlling for participants' baseline belief in vaccines ($ps < .002$; see Tables S16 – S18).

Study 2 revealed that a description of a scientist as having characteristics reflective of higher IH (vs. lower IH) led to greater perceived trustworthiness of the scientist and belief in her scientific research. A description of a scientist with lower IH also led to lower perceived trustworthiness and belief in research than the control scientist, implying that low IH in scientists might be especially detrimental. However, because we described a woman scientist, we were not able to examine whether the effects were in part driven by the scientist's gender. Prior work has argued that the positive outcomes of expressing humility might differ based on social power, such that expressing humility may be less beneficial or even harmful for people of lower social power^{61,62,63}. In particular, others have argued and shown that humility may be harmful if its expression does not adhere to gender norms^{61,64}. In addition, there is evidence that people tend to be biased against women in science. For example, faculty rate women candidates for a STEM position as less competent and less hireable than men candidates^{65,66}. It is therefore possible that low IH was viewed less positively in this study simply because it was perceived as counterstereotypic for a woman scientist. We therefore examined the effect of scientist gender in Study 3.

Study 3

In Study 3 we aimed to test the effect of scientist gender and replicate the results of Study 2 in a new scientific context. We randomly assigned participants ($N = 369$) to read an article about an ostensible psychological scientist researching why people should talk across political divides. Like in Study 2, there was a control condition as well as conditions describing the scientist as high IH or low in IH. We crossed this with the described gender of the scientist (woman: Dr. Sandra Wilson, she/her pronouns; man: Dr. Robert Wilson, he/him pronouns).

We found a large effect of IH condition on perceived IH of the scientist, $F(2, 363) = 167.11$, $p < .001$, $\eta^2_p = .48$, 95% CI for $\eta^2_p = (.41, .54)$. Using a Tukey test to adjust for multiple comparisons, participants saw the scientist as significantly more intellectually humble in the high IH condition ($M = 4.30$, $SD = 0.50$), than in either of the other conditions ($M_{\text{Control condition}} = 3.81$, $SD_{\text{Control condition}} = 0.55$; $M_{\text{Low IH condition}} = 2.96$, $SD_{\text{Low IH condition}} = 0.71$; p 's $< .001$). Participants saw the scientist as significantly less intellectually humble in the low IH condition than in the control condition ($p < .001$). There was an absence of evidence for an effect of ($p = .129$) or interaction with ($p = .548$) the gender of the scientist. The Bayes factor indicated evidence in favor of the null model that only included IH condition (model with gender $BF_{01} = 3.46$; model with gender and its interaction $BF_{01} = 38.02$). Again, the strength of the effect of condition varied by IH subscale (lack of intellectual overconfidence $\eta^2_p = .53$, 95% CI for $\eta^2_p = (.46, .58)$; openness to revising one's views $\eta^2_p = .48$, 95% CI for $\eta^2_p = (.41, .54)$; respect for other's views $\eta^2_p = .28$, 95% CI for $\eta^2_p = (.21, .35)$; independence of intellect and ego $\eta^2_p = .16$, 95% CI for $\eta^2_p = (.09, .22)$).

We found a large effect of IH condition on perceived trustworthiness, $F(2, 363) = 52.70$, $p < .001$, $\eta^2_p = .23$, 95% CI for $\eta^2_p = (.15, .29)$. Participants saw the scientist as significantly more trustworthy in the high IH condition ($M = 6.46$, $SD = 0.69$), than in either of the other conditions ($M_{\text{Control condition}} = 5.98$, $SD_{\text{Control condition}} = 0.81$; $M_{\text{Low IH condition}} = 5.33$, $SD_{\text{Low IH condition}} = 1.13$; p 's $< .001$). Participants saw the scientist as significantly less trustworthy in the low IH condition than in the control condition ($p < .001$). There was an absence of evidence for an effect of ($p = .165$) or interaction with ($p = .957$) the gender of the scientist. The Bayes factor indicated evidence in favor of the null model that only included IH condition (model with gender $BF_{01} = 3.22$, model with gender and its interaction $BF_{01} = 57.45$). Again, the strength of the effect of condition differed by perceived trustworthiness subscale (integrity $\eta^2_p = .23$, 95% CI for $\eta^2_p = (.16, .30)$; benevolence $\eta^2_p = .26$, 95% CI for $\eta^2_p = (.19, .33)$; expertise $\eta^2_p = .14$, 95% CI for $\eta^2_p = (.08, .20)$).

We found a small-to-medium effect of IH condition on belief in the scientist's research on the benefits of talking across political divides, $F(2, 363) = 3.63$, $p = .028$, $\eta^2_p = .02$, 95% CI for $\eta^2_p = (.00, .05)$. Participants believed the scientist's research more in the high IH condition ($M = 4.54$, $SD = 1.07$), than in the low IH condition ($M = 4.17$, $SD = 1.16$; $p = .020$). There was an absence of evidence for the difference between the high IH and control conditions ($M = 4.36$, $SD = 0.97$; $p = .401$). The Bayes factor also indicated some evidence in favor of the null (that group means do not differ; $BF_{01} = 3.06$). There was also an absence of evidence for the

1 difference between the low IH and control conditions ($p = .392$). The Bayes factor also indicated
 2 some evidence in favor of the null ($BF_{01} = 3.04$). Finally, there was an absence of evidence for an
 3 effect of ($p = .414$) or interaction with ($p = .930$) the gender of the scientist. The Bayes factor
 4 indicated evidence in favor of the null model that only included IH condition (model with gender
 5 $BF_{01} = 6.59$, model with gender and its interaction $BF_{01} = 110.31$).

6 Finally, we found a small-medium effect of IH condition on intention to follow the
 7 scientist's research, $F(2, 363) = 5.71$, $p = .004$, $\eta^2_p = .03$, 95% CI for $\eta^2_p = (.00, .07)$. Participants
 8 indicated higher intention to follow the scientist's research in the high IH condition ($M = 5.03$,
 9 $SD = 1.69$), than in the low IH condition ($M = 4.29$, $SD = 1.96$; $p = .003$). There was an absence
 10 of evidence for a difference between the high IH and control conditions ($M = 4.50$, $SD = 1.83$; p
 11 $= .058$), and the Bayes factor indicated weak evidence against the null (that group means do not
 12 differ; $BF_{01} = 0.50$). There was also an absence of evidence for a difference between the low IH
 13 and control conditions ($p = .647$). The Bayes factor indicated evidence in favor of the null (BF_{01}
 14 $= 4.85$). There was an absence of evidence for an effect of ($p = .701$) or interaction with ($p =$
 15 $.642$) the gender of the scientist. The Bayes factor indicated evidence in favor of the null model
 16 that only included IH condition (model with gender $BF_{01} = 7.99$, model with gender and its
 17 interaction $BF_{01} = 101.55$).

18 Study 3 replicated the experimental effects of scientists expressing IH in a different
 19 science domain (psychological science) and did not detect a difference by scientist gender.
 20 However, in Studies 1-3 we did not explicitly assess perceptions of the scientist's race/ethnicity.
 21 Prior research has shown that people tend to presume the racial identity of others based on their
 22 implicit stereotypes, and often assume that those in leadership positions are white⁶⁷. It is
 23 therefore possible that participants in Studies 2 and 3 presumed that the presented scientists were
 24 white, absent of explicit cues. While it is unknown whether there are differential benefits of
 25 perceived IH across racial identities, there is substantial evidence for bias against scientists of
 26 color. For example, faculty rate Black and Latinx candidates as less competent and hireable for
 27 STEM positions⁶⁵. It is therefore possible that scientists of color who exhibit high (vs. low) IH
 28 might be viewed as less confident in their knowledge or less competent. To ensure that the
 29 benefits of perceived IH generalize to scientists of color, in Study 4 we tested if the effects of IH
 30 were affected by the racial identity of the scientist.

31 **Study 4**

32 Study 4 aimed to replicate the results of Studies 2 and 3 in a new scientific context, while
 33 also experimentally varying the described race/ethnicity of the scientist. We randomly assigned
 34 participants ($N = 371$) to read an article about an ostensible climate scientist researching the
 35 benefits of a plant-rich diet for reducing global carbon emissions. We again described the
 36 scientist as either high or low in IH, dropping the neutral control condition to increase power. We
 37 crossed the manipulation of IH with the described race/ethnicity of the scientist using a name
 38 manipulation from prior work (white: Dr. Claire Miller; Black: Dr. Shanice Banks; Latinx: Dr.
 39 Maria Rodriguez)⁶⁵.

We found a large effect of IH condition on perceived IH of the scientist, $F(1, 365) = 468.63$, $p < .001$, $\eta^2_p = .56$, 95% CI for $\eta^2_p = (.50, .61)$. Participants saw the scientist as significantly more intellectually humble in the high IH condition ($M = 4.08$, $SD = 0.61$), than in the low IH condition ($M = 2.64$, $SD = 0.69$). There was an absence of evidence for an effect of ($p = .594$) or interaction with ($p = .223$) the race/ethnicity of the scientist. Further, the Bayes factor indicated strong evidence in favor of the null model that only included IH condition (model with race/ethnicity $BF_{01} = 19.88$, model with race/ethnicity and its interaction $BF_{01} = 95.88$). Again, the strength of the effect of condition varied by IH subscale (lack of intellectual overconfidence $\eta^2_p = .56$, 95% CI for $\eta^2_p = (.50, .61)$; openness to revising one's views $\eta^2_p = .51$, 95% CI for $\eta^2_p = (.44, .57)$; respect for other's views $\eta^2_p = .43$, 95% CI for $\eta^2_p = (.35, .49)$; independence of intellect and ego $\eta^2_p = .30$, 95% CI for $\eta^2_p = (.22, .37)$).

We found a large effect of IH condition on perceived trustworthiness, $F(1, 365) = 84.14$, $p < .001$, $\eta^2_p = .19$, 95% CI for $\eta^2_p = (.12, .26)$. Participants saw the scientist as significantly more trustworthy in the high IH condition ($M = 6.28$, $SD = 0.80$), than in the low IH condition ($M = 5.30$, $SD = 1.24$). There was an absence of evidence for an effect of ($p = .539$) or interaction with ($p = .928$) the race/ethnicity of the scientist. Further, the Bayes factor indicated strong evidence in favor of the null model that only included IH condition (model with race/ethnicity $BF_{01} = 19.96$, model with race/ethnicity and its interaction $BF_{01} = 337.36$). Again, the strength of the effect of condition varied by perceived trustworthiness subscale (integrity $\eta^2_p = .20$, 95% CI for $\eta^2_p = (.13, .27)$; benevolence $\eta^2_p = .23$, 95% CI for $\eta^2_p = (.16, .30)$; expertise $\eta^2_p = .10$, 95% CI for $\eta^2_p = (.05, .16)$).

We found a small-medium effect of IH condition on belief in the scientist's research on the benefits of plant-rich diets, $F(1, 365) = 8.88$, $p = .003$, $\eta^2_p = .02$, 95% CI for $\eta^2_p = (.00, .06)$. Participants believed the scientist's research more in the high IH condition ($M = 4.62$, $SD = 1.35$), than in the low IH condition ($M = 4.17$, $SD = 1.54$). There was an absence of evidence for an effect of ($p = .670$) or interaction with ($p = .722$) the race/ethnicity of the scientist. The Bayes factor indicated evidence in favor of the null model that only included IH condition (model with race/ethnicity $BF_{01} = 23.47$, model with race/ethnicity and its interaction $BF_{01} = 316.90$).

Finally, we found a small-medium effect of IH condition on participants' information seeking behavior, $OR = 1.46$, 95% CI for $OR = (1.16, 1.86)$, $p = .002$, such that a higher percentage of people chose to receive information about switching to a plant-rich diet in the high IH condition (36%) than in the low condition (21%). There was an absence of evidence for an effect of or interaction with the race/ethnicity of the scientist ($ps > .312$).

Study 4 provided additional evidence of the benefits of expressed IH in a different (and politically polarized) science domain (climate science) and also across scientist races/ethnicities.

Study 5

Studies 1-4 showed that seeing a scientist as more intellectually humble increases perceived trustworthiness of them and their findings compared to seeing them as less intellectually humble. These results raise an important question: how can scientists most effectively express their IH? To answer this question, we first reached out to experts in the field

of IH to identify ways that scientists can communicate with IH to the public. We organized the most commonly reported communication approaches into three categories: personal intellectual humility (e.g., updating beliefs; willingness to share credit with others), limits of methods (e.g., describing specific methodological limitations; methodological constraints of the current research), and limits of results (e.g., describing the nuances and weaknesses in the results of the current research; describing limits to the generalizability of the findings). In Study 5, we randomly assigned participants from a census-matched sample ($N = 679$), to read one of four (three IH communication conditions, one neutral control) interviews with a scientist discussing the psychological benefits of taking a break from social media⁶⁸. Critically, this was a subtle manipulation of communicating IH (vs. a neutral control) about a non-polarizing topic.

We found a small-to-medium effect on perceived IH of the scientist, $F(3, 675) = 8.93$, $p < .001$, $\eta^2_p = .04$, 95% CI for $\eta^2_p = (.01, .07)$ (see Figure 4). Using a Tukey test, participants saw the scientist as more intellectually humble in the personal intellectual humility condition ($M = 3.97$, $SD = 0.47$) and the limitations of results condition ($M = 3.93$, $SD = 0.50$) than the control condition ($M = 3.70$, $SD = 0.53$, $ps < .001$). Perceived IH was also significantly higher in the personal intellectual humility condition than the limitations of methods condition ($M = 3.82$, $SD = 0.55$, $p = .033$). There was an absence of evidence for a difference between the limitations of methods condition and the control ($p = .173$). The Bayes factor indicated the data was insensitive to detect this difference ($BF_{01} = 1.38$). The effect of condition was very similar across all subscales ($\eta^2_p = .02-.03$).

Counter to our expectations, there was no effect of condition on perceived trustworthiness, $F(3, 675) = 2.59$, $p = .052$, $\eta^2_p = .01$, 95% CI for $\eta^2_p = (.00, .03)$. The Bayes factor also indicated evidence in support for the null hypothesis ($BF_{01} = 5.45$). This may be due to a ceiling effect, with all conditions having a mean of over 6 on a 7-point scale ($M_{\text{control}} = 6.05$, $SD_{\text{control}} = 0.83$; $M_{\text{limits of methods}} = 6.15$, $SD_{\text{limits of methods}} = 0.79$; $M_{\text{limits of results}} = 6.28$, $SD_{\text{limits of results}} = 0.65$; $M_{\text{personal intellectual humility}} = 6.17$, $SD_{\text{personal intellectual humility}} = 0.77$). Interestingly, the effect of condition was significant when looking at the individual subscales of integrity, $F(3, 675) = 2.81$, $p = .039$, $\eta^2_p = .01$, 95% CI for $\eta^2_p = (.00, .03)$, and benevolence, $F(3, 675) = 3.27$, $p = .021$, $\eta^2_p = .01$, 95% CI for $\eta^2_p = (.00, .03)$, but not expertise, $F(3, 675) = 1.30$, $p = .274$, $\eta^2_p = .01$, 95% CI for $\eta^2_p = (.00, .02)$. In a secondary mediation analysis, we found a significant indirect effect on perceived trustworthiness from both limits of results ($b = 0.18$, $SE = 0.05$, 95% CI for $b = (0.09, 0.28)$) and personal intellectual humility ($b = 0.22$, $SE = 0.05$, 95% CI for $b = (0.13, 0.31)$) conditions through perceived IH using 10,000 bootstrapped samples (compared to the control). There was no indirect effect from the limits of methods condition ($b = 0.09$, $SE = 0.05$, 95% CI for $b = (-0.00, 0.19)$; see Figure S9).

We found a small-to-medium effect of IH condition on belief in the scientist's research on taking a break from social media, $F(3, 675) = 9.26$, $p < .001$, $\eta^2_p = .04$, 95% CI for $\eta^2_p = (.01, .07)$. Using a Tukey test, participants believed the research more in the personal intellectual humility condition ($M = 5.32$, $SD = 0.97$) than in the limits of results condition ($M = 4.89$, $SD = 1.07$, $p < .001$) and the limits of methods condition ($M = 5.02$, $SD = 1.08$, $p = .033$). Participants

believed the research more in the control condition ($M = 5.39$, $SD = 0.97$) than in the limits of results condition ($p < .001$) and the limits of methods condition ($p = .005$). The personal intellectual humility condition and the control condition did not differ ($p = .927$). For the latter contrast, the Bayes factor also indicated support for the null hypothesis ($BF_{01} = 6.69$). We ran the same mediation model on belief in the scientists' research and found significant indirect effects from both limits of results ($b = 0.10$, $SE = 0.03$, 95% CI for $b = (0.05, 0.17)$) and personal intellectual humility ($b = 0.13$, $SE = 0.03$, 95% CI for $b = (0.07, 0.20)$) conditions (compared to the control) through perceived IH. There was no indirect effect from the limits of methods condition ($b = 0.05$, $SE = 0.03$, 95% CI for $b = (-0.00, 0.11)$).

Finally, we found non-significant effects on participants' information seeking behavior relative to the control (personal intellectual humility condition $OR = 0.65$, 95% CI for $b = (0.40, 1.04)$, $p = .073$; limitations of methods condition $OR = 0.67$, 95% CI for $b = (0.42, 1.07)$, $p = .093$; limitations of results condition $OR = 0.82$, 95% CI for $b = (0.52, 1.28)$, $p = .383$).

Study 5 provided mixed evidence for the effectiveness of the expert-recommended approaches to communicating IH. We found that two of the three approaches effectively increased perceptions of IH of the ostensible scientist (personal intellectual humility and limitations of results). Further, perceptions of IH significantly predicted perceived trustworthiness as in prior studies ($r(677) = 0.56$, $p < .001$). However, only personal intellectual humility successfully increased perceptions of IH without backfiring, and even this approach had only indirect effects on perceived trustworthiness and belief in the scientist's research.

Discussion

In the current research, we aimed to test the effects of seeing scientists as intellectually humble on trust in scientists and intentions to follow scientific evidence. In line with our predictions and prior theorizing³³, we found that perceiving scientists as high in IH is reliably associated with perceiving them as trustworthy and believing their research (Study 1). We also tested this experimentally and found that describing a scientist as high (vs. low) in IH increases perceptions of their trustworthiness and belief in their research (Studies 2-4). Critically, this effect was consistent across scientists with different gender (Study 2) and racial/ethnic (Study 3) identities. We also found that perceptions of IH contribute to more perceived trustworthiness in scientists across different topics including medical, climate, and psychological science. Finally, we found that perceptions of IH contribute to intentions to follow the scientist's research-based recommendations and information-seeking behavior aimed at learning more about the scientist's recommendations. Together, these four studies offer compelling evidence that perceptions of scientists' IH play an important role in both trust in scientists and willingness to follow their research-based recommendations.

In Study 5, we tested three communication approaches to increase perceptions of IH and perceived trustworthiness. Of the approaches tested, personal intellectual humility was the most successful. This is perhaps unsurprising, as people might pay closest attention to expressions that reflect a scientist's personal character (rather than limitations of the research) when making judgments of their trustworthiness. However, even this approach was limited in its direct benefits

on perceived trustworthiness and intentions to follow scientific evidence, perhaps due in part to its subtlety and its comparison to a neutral (rather than low IH) control. Future work is needed to provide a deeper understanding of how scientists can authentically express IH when communicating about their science with the public and whether such communication enhances perceived trustworthiness.

We would be remiss to not acknowledge that two of the IH communication approaches backfired in the domain of believing a scientist's research. This finding raises the question of how perceived trustworthiness in the scientist as a person might differ from trust in the scientist's results. It is possible that conveying limitations about a particular study might signal that the scientist is more honest and intellectually humble but might simultaneously reduce trust in the research or recommendations that follow from it⁶⁹. This is perhaps why prior research has found mixed effects of expressing limitations of research^{41,43,44,70}. Future research should therefore seek to understand other ways to convey IH, while paying close attention to potential backfire effects. If certain expressions of IH are found to be helpful in increasing trust without decreasing belief in the science, it may be important to consider how to enhance the culture of intellectually humble communication within scientific institutions as well as between scientists and the public.

This research was characterized by several strengths distinguishing it from other studies in this domain (e.g., use of experimental design, crowdsourcing of IH researchers, behavioral information-seeking measures), and it is, to our knowledge, the first set of studies to examine how participants perceive IH in scientists. Nevertheless, this research is also characterized by several important limitations. First, all studies used convenience samples collected from online pools. While this does make them more diverse than typical university pools⁷¹, it also limits their generalizability (e.g., all participants chose to opt in to academic research websites and therefore may have more positive attitudes towards science). Similarly, while we collected a census matched sample in Study 5, the samples for Studies 1-4 were notably less diverse in terms of race/ethnicity. Second, this research did not test if there are specific underlying mechanisms for why perceptions of IH increase perceived trustworthiness. It will be fruitful to identify what precisely it is about IH that leads to more trust, and whether these effects are unique to IH as opposed to other correlated traits/processes (e.g., prosociality, agreeableness; see Figures S3, S5, S8). Future research should also directly test if perceived IH combats common drivers of lower perceived trustworthiness, such as conspiracy beliefs^{17,21} or perceptions of ideological motives¹⁹. It would also be advantageous to examine whether seeing a scientist model IH boosts the perceiver's IH, thereby motivating people to seek out accurate scientific knowledge themselves. Third, while we chose scientific disciplines that vary in their methods and topics of study, the current research can only speak to the effect of IH in these specific disciplines (medical science, climate science, psychological science). Future research should replicate this work in other disciplines (e.g., economics, engineering, political science), as such work would clarify the generalizability of the relations identified in the present investigation. Finally, we used samples from the United States and cannot speak to how these results generalize to other geographic or cultural contexts.

In conclusion, this research builds on an emerging literature revealing the importance of IH in helping to solve major societal problems. IH is undoubtedly critical to the scientific process³³. The current studies are some of the first to investigate how perceptions of people's IH impact their judgments, and we examined this question in a context that has paramount consequences for the wellbeing of society and its members: perceived trustworthiness of scientists and willingness to follow their research-based recommendations. Our work suggests that participants understand that IH is a beneficial characteristic for scientists to possess in these disciplines. At a practical level, this work suggests that scientists embracing and conveying IH is beneficial. Across three studies in different scientific domains, we found that scientists who are described as possessing intellectually humble characteristics—such as being willing to admit gaps in their knowledge, listening to the input of others, and revising their beliefs in the face of new evidence—were more trusted than scientists who were described as possessing characteristics consistent with lower IH. Moreover, in Study 5, we found that expressions of personal intellectual humility (e.g., admitting that one's original predictions were wrong and updating one's thinking) promote perceptions of scientist IH, with downstream benefits for perceived trustworthiness of the scientist and their research. Although more work is needed to identify specific communication strategies that scientists can use to effectively convey IH, the current set of studies speaks to the benefits of scientists approaching their science and their public communications in ways that reflect high rather than low IH. In particular, given that we saw few differences between the high IH and neutral control conditions in Studies 2 and 3, this may suggest that the default perception of scientists is that they possess relatively high levels of IH and may especially dislike it when scientists display low IH or even arrogance. While future research should continue to disentangle valuing IH from disvaluing intellectual overconfidence or arrogance, we encourage scientists to be particularly mindful of displaying low IH, such as by expressing overconfidence, being unwilling to course-correct, or disrespecting others' views. These results also suggest that the field of science might more deliberately discourage low IH practices and reward scientists who approach their research and scientific communication with IH (as suggested by³³).

Methods

Ethics Statement

All studies were approved by the University of Pittsburgh IRB and determined to be exempt from requiring consent. All participants for each study read an information script prior to participating that informed them of the procedure, risks/benefits, and compensation.

Study 1

The goal of Study 1 was to examine the public's perception of scientists' IH, and to test what these perceptions predict. Study 1 was preregistered on June 5, 2022, at: https://osf.io/nwbc2/?view_only=c10fa8e609444bf19bb58043c0daf95c. This and all studies were conducted in Qualtrics.

Participants

An a priori power analysis in G*Power⁷² recommended a sample of 134 to detect a medium effect ($r = .30$)⁶⁰ using a correlational design (95% power, $\alpha = .05$, two-tailed). We aimed to collect data from 300 participants to allow for tests for moderation by political orientation. We collected a sample of 300 participants from Prolific Academic. Of these, two failed our preregistered attention check, leaving a sample of 298 (47.65% Female, 49.66% Male, 2.35% Non-binary/third gender, 0.34% Prefer not to say; 2.01% Native American, 9.73% Asian or Asian American, 1.68% Black or African American, 0.67% Native Hawaiian or other Pacific Islander, 10.07% Hispanic or Latina/Latino/Latinx, 84.90% White, 2.01% Middle Eastern or North African; $M_{age} = 39.40$, $SD_{age} = 14.78$). We collected equal numbers of liberals and conservatives in the sample ($Ns = 149$).

Materials and Procedure

Perceived IH of scientists. We instructed participants to “consider a broad range of scientists, such as medical scientists (e.g., scientists who study human health conditions and diseases and how to treat them), social scientists (e.g., psychologists; archaeologists), and natural scientists (e.g., physicists; biologists).” Participants then completed an adapted version of the Comprehensive Intellectual Humility Scale⁵⁵. This is a validated 22-item measure containing four subscales—*independence of intellect and ego* (e.g., “When someone disagrees with ideas that are important to a scientist, it makes the scientist feel insignificant”), *openness to revising one’s viewpoint* (“Scientists are open to revising their important beliefs in the face of new information”), *respect for others’ viewpoints* (“Scientists respect that there are ways of making important decisions that are different from the way they make decisions”) and *lack of intellectual overconfidence* (“Scientists believe that their ideas are usually better than other people’s ideas”)—on a scale from 1 (Strongly disagree) to 5 (Strongly agree; $\alpha = .94$). See Table S7 for results broken up by subscales across all studies.

Perceived trustworthiness of scientists. Participants then indicated their perceived trustworthiness of scientists on the Muenster Epistemic Trustworthiness Inventory³². This is a validated measure containing three subscales of 14 bipolar items—*expertise* (e.g., *incompetent – competent*), *integrity* (e.g., *unjust – just*), and *benevolence* (e.g., *immoral – moral*)—on seven-point scales ($\alpha = .95$). See Table S8 for results broken up by subscales across all studies.

Belief in polarizing science topics. Participants next completed three scales assessing beliefs in polarizing science topics in a randomized order. Participants completed five-item measures assessing each belief in climate change (e.g., “Human CO2 emissions cause climate change”; $\alpha = .94$), support for vaccinations (e.g., “I believe that vaccines are a safe and reliable way to help avert the spread of preventable diseases”; $\alpha = .91$) and support for genetically modified foods (e.g., “I believe that genetic modification is an important and viable contribution to help feed the world’s rapidly growing population”; $\alpha = .93$) on scales from 1 (Strongly disagree) to 7 (Strongly agree)²¹.

Exploratory measures. Participants completed exploratory measures of conspiracist ideation⁷² ($\alpha = .87$), science knowledge⁷⁴, cognitive reflection⁷⁵, and participants’ own IH⁵⁵ ($\alpha = .88$). Finally, participants completed political (political identity, political orientation, political conviction, political interest, moralization of politics), science-related (interest in science, views

of scientists in politics), and demographic (race/ethnicity, age, gender, education, religion, religiosity) measures. See Tables S2, S3-S6 for analyses with exploratory measures.

Study 2

The goal of Study 2 was to examine the experimental impacts of scientist intellectual humility on perceived trustworthiness. Study 2 was preregistered on August 8, 2022, at:

https://osf.io/26fjk/?view_only=fa43c2b8c7694afc925e9d1206260b51.

Participants

An a priori power analysis in G*Power⁷² recommended a sample of 252 to detect a medium effect ($f = .25$)⁶⁰ using a three-condition design (95% power, $\alpha = .05$). We aimed to double this and collect 520 to test for moderation by political orientation. We collected a sample of 517 from Prolific Academic. Of these, 152 failed the first attention check, 9 failed the second attention check, and 76 failed the manipulation check, leaving a sample of 317 (47.95% Female, 49.53% Male, 2.21% Non-binary/third gender, 0.32% Prefer to self-describe; 1.58% Native American, 8.20% Asian or Asian American, 6.31% Black or African American, 0.32% Native Hawaiian or other Pacific Islander, 10.41% Hispanic or Latina/Latino/Latinx, 78.55% White, 0.63% Middle Eastern or North African, 0.95% “Other”; $M_{\text{age}} = 40.05$, $SD_{\text{age}} = 15.10$). We collected roughly equal numbers of liberals ($N = 160$) and conservatives ($N = 157$).

Materials and Procedure

Manipulation. We randomly assigned participants to one of three conditions. All participants read an article about an ostensible scientist (Dr. Susan Moore) at the University of Michigan researching a new anti-viral treatment for long COVID-19 symptoms. We described Dr. Moore as high in IH (e.g., “...according to one colleague, ‘Dr. Moore is not afraid to admit when she doesn’t yet know something.’” $N = 115$), low in IH (e.g., “...according to one colleague, ‘Dr. Moore is not afraid to assert what she knows.’” $N = 103$), or no information about IH ($N = 99$). We modeled the manipulations using the IH behaviors expressed in the Comprehensive Intellectual Humility Scale.

Big Five Inventory of the Scientist. As a filler scale, we asked participants to report on their perceptions of the scientist’s personality⁷⁶ using a 15-item measure with subscales of extraversion (e.g., “Dr. Moore is someone who is dominant, acts as a leader.”; $\alpha = .57$), openness (e.g., “Dr. Moore is someone who is original, comes up with new ideas.”; $\alpha = .61$), neuroticism (e.g., “Dr. Moore is someone who worries a lot.”; $\alpha = .54$), conscientiousness (e.g., “Dr. Moore is someone who is reliable, can always be counted on”; $\alpha = .65$), and agreeableness (e.g., “Dr. Moore is compassionate, has a soft heart”; $\alpha = .81$) on a scale from 1 (Strongly disagree) to 5 (Strongly agree; see Tables S9, S19-S21).

Perceived IH of the Scientist. As a manipulation check, participants completed the same IH scale from Study 1, now adapted to be about Dr. Moore (e.g., “Dr. Moore believes that her ideas are usually better than other people’s ideas”; $\alpha = .97$).

Likability of the Scientist. Participants also completed an adapted four-item likability scale⁷⁷ (e.g., “Dr. Moore is friendly”) on a scale from 1 (Very strongly disagree) to 7 (Very strongly agree; $\alpha = .94$; see Tables S9, S13-S15).

Perceived Trustworthiness of the Scientist. Participants completed the same Muenster Epistemic Trustworthiness Inventory, now focused on Dr. Moore ($\alpha = .96$).

Belief in the Scientist's Research. Participants completed a five-item measure assessing belief in the scientist's anti-viral treatment (e.g., "I believe that this new anti-viral treatment is a safe and reliable way to help avert long COVID-19 symptoms"; adapted from²¹; $\alpha = .88$).

Exploratory measures. As an exploratory covariate participants completed the same support for vaccinations scale ($\alpha = .94$) and interest in science scale (inter-item $r(315) = .39$) as in Study 1. Participants completed the same demographic measures as in Study 1. See Table S9 and Tables S16-S18 for analyses on exploratory measures.

Study 3

The goal of Study 3 was to replicate the results of Study 2, while also testing the impacts of scientist gender on perceived IH. Study 3 was preregistered on March 8, 2023, at:

https://osf.io/m4xzv/?view_only=0090326896de4db2b869ce46fbe16c1f.

Participants

An a priori power analysis in G*Power⁷² recommended a sample of 374 to detect a small-medium effect ($f = .20$)⁶⁰ using a 3 X 2 between-subjects design (95% power, $\alpha = .05$). We collected a sample of 450 from Prolific Academic to account for potential low-quality data. Of these, 3 failed the first attention check, 79 failed the second attention check, leaving a sample of 369 (49.86% Female, 48.24% Male, 1.90% Non-binary/third gender; 2.17% Native American, 5.96% Asian or Asian American, 7.05% Black or African American, 0.54% Native Hawaiian or other Pacific Islander, 8.67% Hispanic or Latina/Latino/Latinx, 82.66% White, 0.27% Middle Eastern or North African, 0.27% "Other"; $M_{age} = 42.63$, $SD_{age} = 14.40$). We collected roughly equal numbers of liberals ($N = 195$) and conservatives ($N = 174$).

Materials and Procedure

Manipulation. Like in Study 2, we randomly assigned participants to one of three IH conditions (high, $N = 146$; low, $N = 113$; control, $N = 110$). The IH part of the manipulation was identical to Study 2. Unlike Study 2, all articles were about an ostensible psychological scientist at the University of Pittsburgh researching the benefits of talking to people across the political divide in the United States. We crossed IH condition with the gender of the scientist, who was either a woman (Dr. Sandra Wilson, $N = 184$) or a man (Dr. Robert Wilson, $N = 185$).

Perceived IH of the Scientist. Participants completed the same IH scale from Studies 1 and 2, now adapted to be about Dr. Wilson ($\alpha = .96$).

Perceived Trustworthiness in the Scientist. Participants completed the Muenster Epistemic Trustworthiness Inventory about the scientist ($\alpha = .97$).

Belief in the Scientist's Research. Participants completed a five-item measure assessing belief in the scientist's research on the benefits of talking across the political divide (e.g., "I believe that talking to people from the opposing political party makes people feel happier;" adapted from²¹; $\alpha = .72$).

Intention to Follow Research. Participants completed two items about their intentions to learn more about how to follow the scientist's research recommendations ("I would be interested

in receiving information about how to have productive conversations with people from other political parties,” “I would be interested in learning more about Dr. Wilson’s suggestions for productive conversations;” inter-item $r(366) = .86$).

Stereotype Content of the Scientist. As an exploratory outcome, participants completed two items assessing the warmth (“warm,” “friendly;” inter-item $r(366) = .91$) and two items assessing the competence (“competent,” “capable;” inter-item $r(365) = .88$) of the scientist (adapted from⁷⁸; see Tables S36-S50 for results). It should be noted that the “competent” item from the competence scale overlaps with one item from the Muenster Epistemic Trustworthiness Inventory (incompetent—competent). We therefore recommend caution when interpreting models that include both these scales.

Demographic measures. Participants completed the same demographic measures as in prior studies.

Study 4

The goal of Study 4 was to replicate the results of Studies 2 and 3 and ensure the effects of perceived IH were consistent across scientists with different race/ethnic identities. Study 4 was preregistered on June 15, 2023, at:

https://osf.io/yjdf2/?view_only=c413801038924d2fa81ec177077254f1.

Participants

An a priori power analysis in G*Power⁷² recommended a sample of 374 to detect a small-medium effect ($f = .20$)⁶⁰ using a 3 X 2 between participant design (95% power, $\alpha = .05$). We collected a sample of 450 from Prolific Academic to account for potential low-quality data. Of these, 9 failed the first attention check, 74 failed the second attention check, leaving a sample of 371 (51.21% Female, 47.17% Male, 1.35% Non-binary/third gender, 0.27% Prefer to self-describe; 0.54% Native American, 8.63% Asian or Asian American, 6.47% Black or African American, 8.63% Hispanic or Latina/Latino/Latinx, 83.02% White, 0.54% Middle Eastern or North African, 0.27% “Other”; $M_{age} = 43.07$, $SD_{age} = 14.85$). We collected roughly equal numbers of liberals ($N = 187$) and conservatives ($N = 184$).

Materials and Procedure

Manipulation. We randomly assigned participants to one of two IH conditions (high, $N = 198$; low, $N = 173$). We removed the neutral control condition to maximize power. The IH part of the manipulation was identical to Studies 2 and 3. Unlike Studies 2 and 3, participants read about an ostensible climate scientist researching why people should switch to plant-rich diets to reduce global emissions. We modeled the reasoning and results off real research done by Project Drawdown (drawdown.org). We crossed IH condition with the race/ethnicity of the scientist. We used a name manipulation from prior work⁶⁵; the scientist was either Dr. Claire Miller (White condition, $N = 133$), Dr. Shanice Banks (Black condition, $N = 116$), or Dr. Maria Rodriguez (Latinx condition, $N = 122$). It should be noted that despite using a manipulation from prior work, 98 participants misidentified the race/ethnicity of the scientist. Removing these participants did not change any of the results with race/ethnicity condition.

Perceived IH of the Scientist. Participants completed the same IH scale from prior studies ($\alpha = .97$).

Perceived Trustworthiness of the scientist. Participants completed the Muenster Epistemic Trustworthiness ($\alpha = .97$).

Belief in the Scientist's Research. Participants completed a four-item measure assessing belief in the scientist's research on the benefits of switching to a plant-rich diet to reduce emissions (e.g., "I believe that adopting a plant-rich diet will reduce climate change;" adapted from ²¹; $\alpha = .83$).

Information Seeking Behavior. Participants had the option to opt into ostensibly being sent information and tips for switching to a plant rich diet with answer options of no (0) and yes (1). Participants also reported on their current level of meat consumption ("How many days per week do you currently eat meat (on average)," "What percentage of your current meals are plant-rich"; data from this measure are available on OSF).

Stereotype Content of the Scientist. Participants completed the same measure of warmth (inter-item $r(369) = .93$) and competence (inter-item $r(368) = .87$) of the scientist (see Table S64 and Tables S69-S78 for results).

Demographic measures. Participants completed the same demographic measures as in prior studies.

Study 5

The goal of Study 5 was to test communication approaches scientists might use to effectively express IH when communicating to the public. Study 5 was preregistered on December 8, 2023, at https://osf.io/mb5sx/?view_only=8e5d50e477fa4d1f9598e32eaf3c5c15.

Participants

An a priori power analysis in G*Power⁷² recommended a sample of 768 to detect a small-medium effect ($f = .20$)⁶⁰ using a four-condition design (95% power, $\alpha = .05$). We collected a census matched sample of 775 from CloudResearch Connect. We removed one participant who completed the survey twice. Following our preregistered exclusion criteria, we removed 66 participants who spent less than five seconds on the manipulation, 25 participants who failed the first attention check, and 13 participants who failed the second attention check. This left a final sample of 679 (49.19% Female, 49.63% Male, 0.29% Non-binary or Transgender; 2.65% Native American, 5.30% Asian or Asian American, 13.11% Black or African American, 0.29% Native Hawaiian or other Pacific Islander, 14.73% Hispanic or Latina/Latino/Latinx, 75.85% White, 0.15% Middle Eastern or North African, 0.88% "Other"; $M_{\text{age}} = 46.51$, $SD_{\text{age}} = 15.94$). We also screened to include a sample that was on average politically moderate ($M_{\text{political orientation}} = 3.86$, $SD_{\text{political orientation}} = 2.06$ on a 7-point scale).

Materials and Procedure

Manipulation. To determine what communication approaches scientists might use to convey IH when describing their research, we reached out to experts in the field of IH via email to ask them how they convey IH to the public, or how they think others might. In all, eleven experts provided a total of 34 recommendations. We then grouped these together into three different

approaches. The first, personal levels of humility, was operationalized as admitting that one's initial predictions were wrong or incorrect/updating beliefs and by giving credit to graduate students. The second, limits of methods, was operationalized as explicitly stating that there are methodological limitations and by acknowledging that one study cannot prove something is universally true. The third, limits of results, was operationalized as acknowledging that there are inconsistent findings and being cautious not to overgeneralize.

We randomly assigned participants to read one of four short interviews with a psychological scientist, Dr. Susan Moore at the University of Pittsburgh, who was researching why it is beneficial to take breaks from social media (modeled off real research by⁶⁸). One condition was a neutral control and the other three conditions contained one of each of the IH communication approaches ($N_{\text{Control}} = 164$, $N_{\text{Limits of methods}} = 174$, $N_{\text{Limits of results}} = 178$, $N_{\text{Personal intellectual humility}} = 163$).

Perceived IH of the Scientist. Participants completed the same IH scale from prior studies ($\alpha = .93$).

Perceived Trustworthiness of the Scientist. Participants completed the Muenster Epistemic Trustworthiness Inventory ($\alpha = .96$).

Belief in the Scientist's Research. Participants completed a four-item measure assessing belief in the scientist's research on the benefits of taking a break from social media (e.g., "I believe taking a break from social media makes people happier and less stressed.," adapted from²¹; $\alpha = .68$).

Information Seeking Behavior. Participants indicated if they would like to be sent information and tips for taking a social media break with answer options of no (0), yes (1), and "I don't use social media." Participants who indicated the latter were removed from analyses on this measure.

Stereotype Content of the Scientist. Participants also completed the same measure of warmth (inter-item $r(676) = .91$) and competence (inter-item $r(675a) = .85$) of the scientist (see Table S92 for results).

Ratings of individual statements. As an exploratory measure, we presented all participants with six different IH statements included in the manipulations. We asked participants how much each statement conveyed IH, would increase their perceived trustworthiness in the scientist, and would increase their trust in the scientist's research (see Figure S11 and Tables S104-S106 for results).

Demographic measures. Participants completed the same demographic measures as in prior studies.

Data Availability

Anonymized data is available at
https://osf.io/d3xua/?view_only=0ebe5366331f497da7c78a7c874c2961.

Code Availability

Code is available at
https://osf.io/d3xua/?view_only=0ebe5366331f497da7c78a7c874c2961.

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Author Contributions Statement

8 JK, KS, SMB, and NV designed the studies. JK and KS collected the data. JK analyzed
9 the data. JK and KS wrote the first draft of the manuscript. NV and SMB provided comments on
10 and edited the manuscript.

Competing Interest Statement

12 The authors have no competing interests to declare.

13
14

Figures

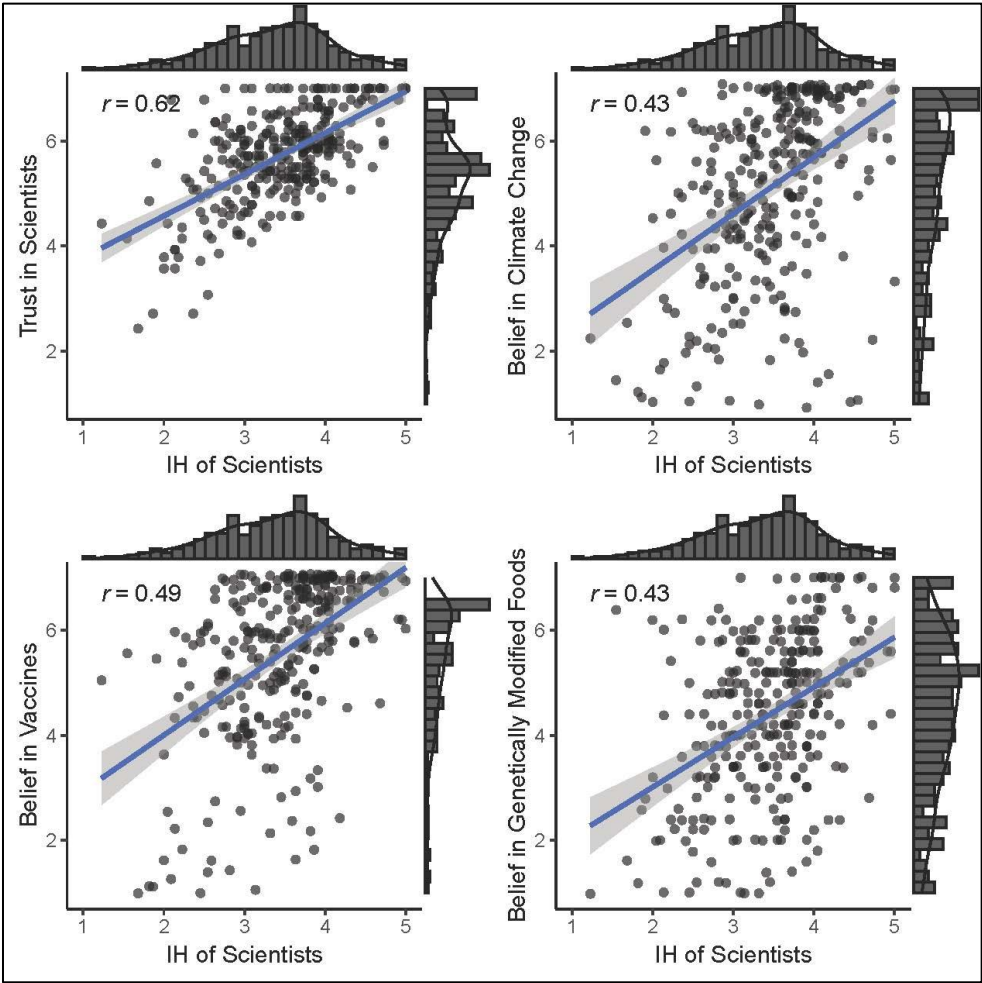


Figure 1. Associations between perceived IH of scientists and outcomes. Error band represents 95% confidence interval. Correlation tests are two-sided. N = 298. Study 1

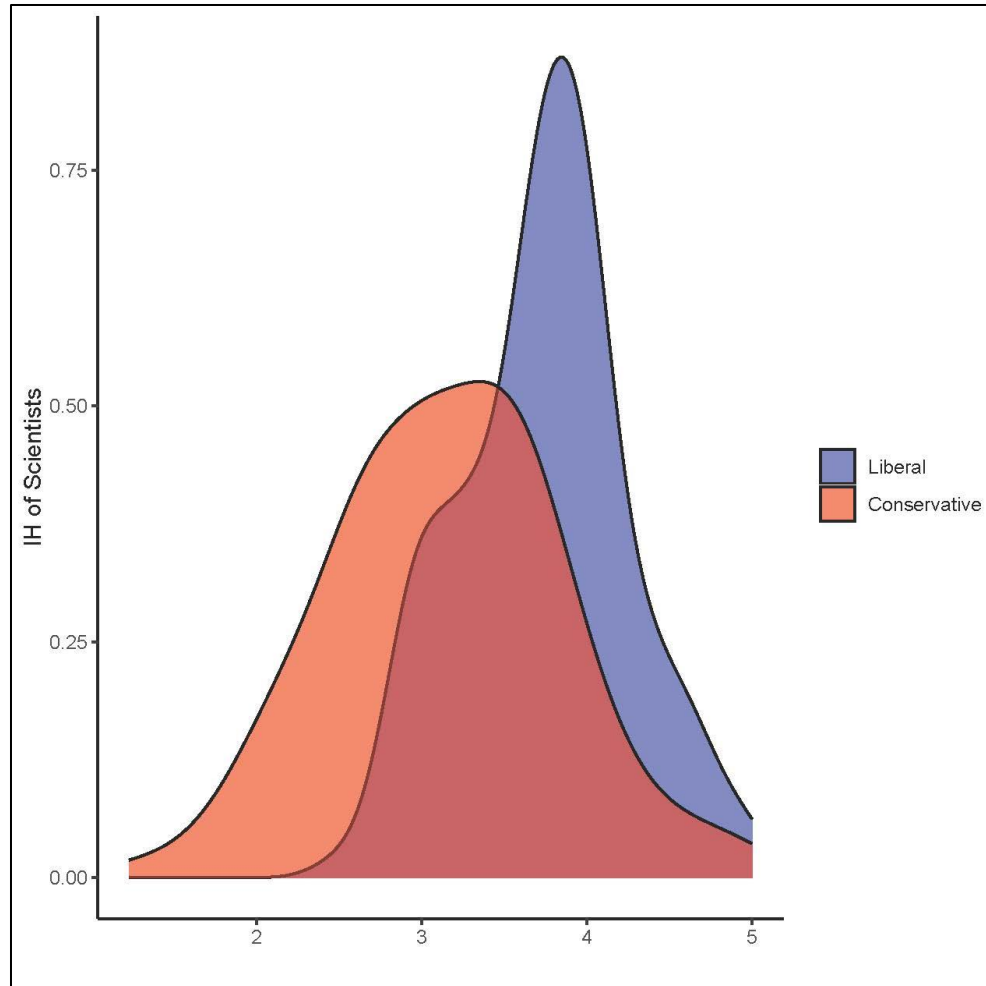


Figure 2. Density plot of perceived IH of scientists separated by political orientation. N = 298.

Study 1

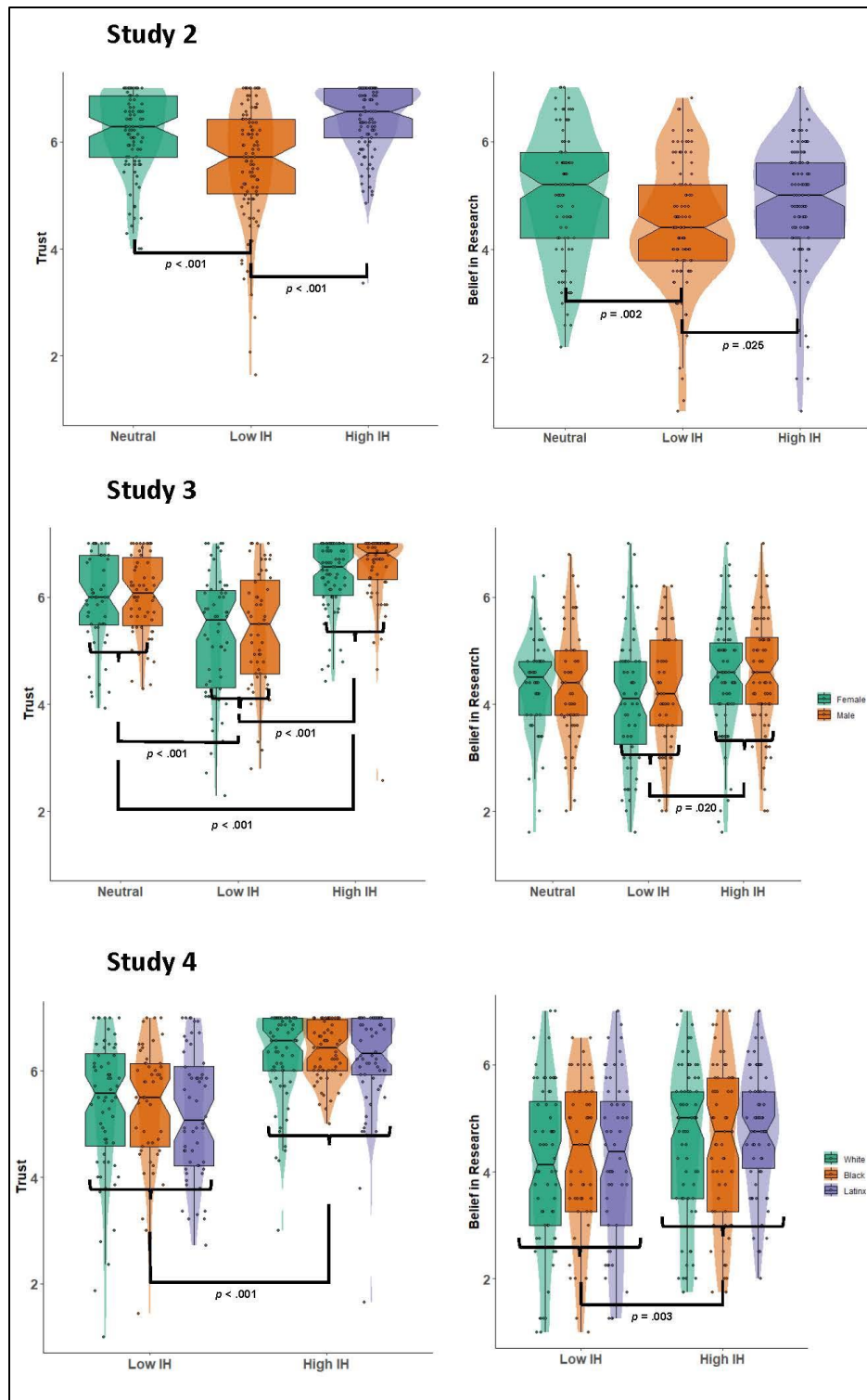


Figure 3. Violin and boxplots displaying differences in perceived trustworthiness (left) and belief in research (right) by condition. Line indicates median value, notches indicate 95% confidence intervals around the median, upper and lower hinges indicate 75th and 25th percentiles respectively, whiskers indicate 1.5*inter-quartile range. All significant contrasts are labeled. $N_{\text{Study 2}} = 317$, $N_{\text{Study 3}} = 369$, $N_{\text{Study 4}} = 371$. Studies 2-4.

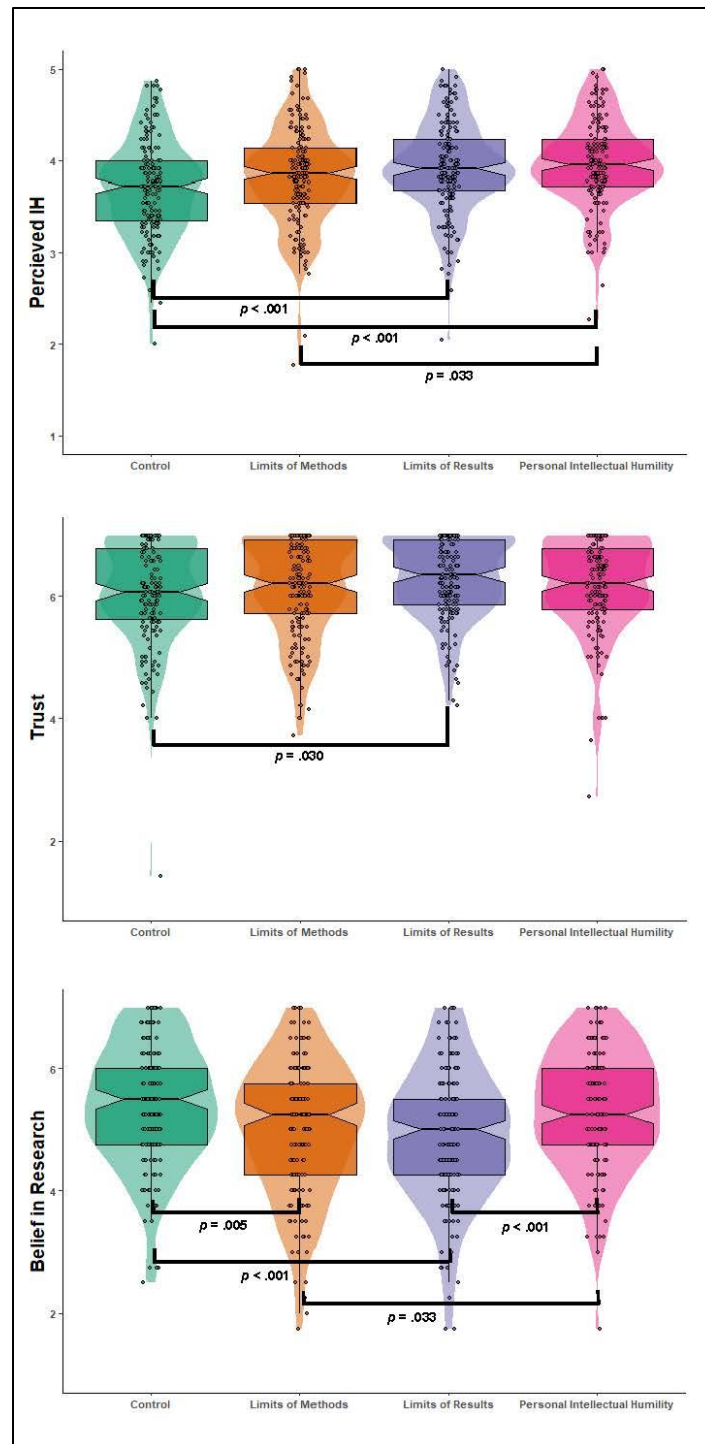


Figure 4. Violin and boxplots displaying differences in perceived IH (top), trust (middle), and belief in research (bottom) by condition. Line indicates median value, notches indicate 95% confidence intervals around the median, upper and lower hinges indicate 75th and 25th percentiles respectively, whiskers indicate 1.5*inter-quartile range. All significant contrasts are labeled. N = 679. Study 5

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